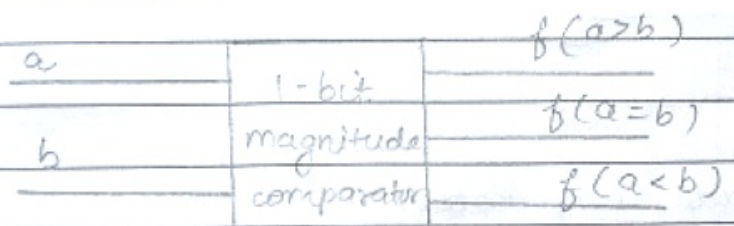


Magnitude Comparator

It is a combinational circuit which compares two binary numbers in order to determine if first number is larger or equal or smaller than second.

1-bit magnitude comparator (Amplitude comparator)



a	b	$f_{a>b}$	$f_{a=b}$	$f_{a<b}$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

Ques. Implement 1-bit mag. comp. using suitable decoder

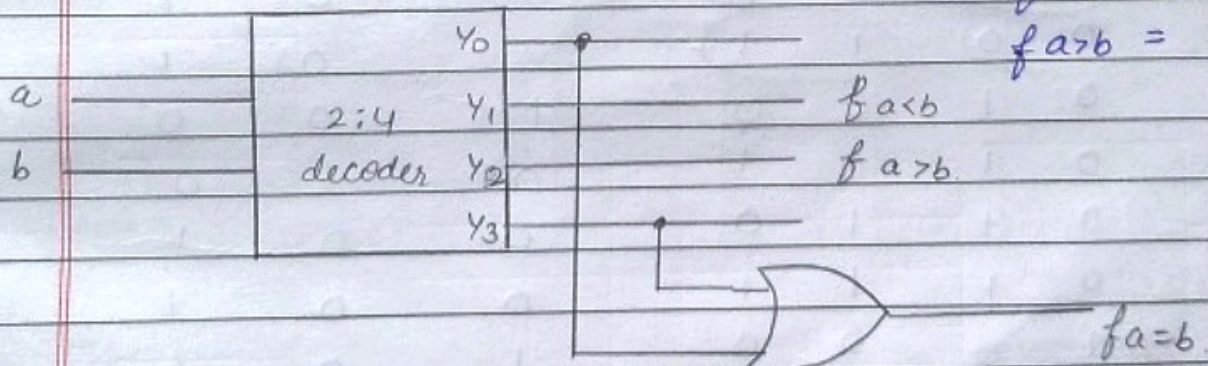
suitable : 2:4

$f_{a=b} \Rightarrow Y_0, Y_3$

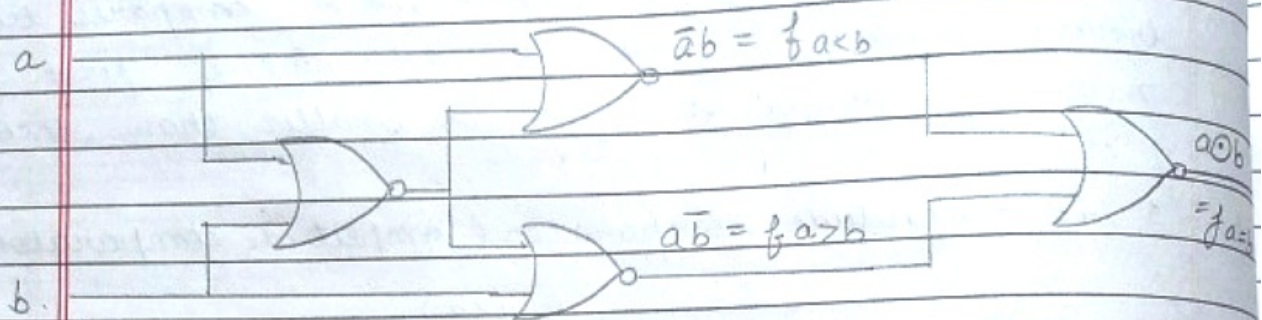
$f_{a=b} = a \odot b$

$f_{a<b} = \bar{a}b$

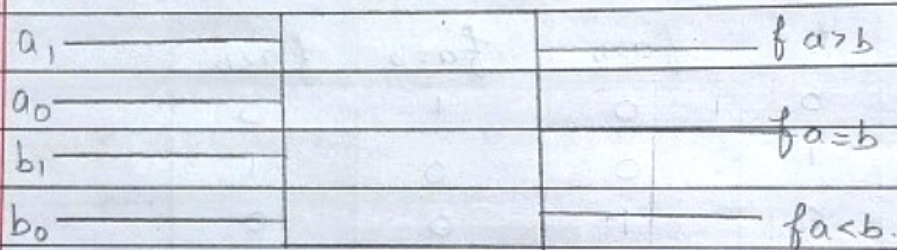
$f_{a>b} = a\bar{b}$



Ques. Implement 1 bit magnitude comparator using minimum no. of NOR gate.



2-bit magnitude comparator.



a_1	a_0	b_1	b_0	$f_{a > b}$	$f_{a = b}$	$f_{a < b}$
0	0	0	0	0	1	0
0	0	0	1	0	0	0
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	0
1	0	1	1	0	0	0
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	0	0	0
1	1	1	1	0	0	0

divisible by 5.

$$f_{a=b} = \sum m(0, 5, 10, 15) \leftarrow$$

$$f_{a>b} = \sum m(4, 8, 9, 12, 13, 14)$$

$$= a_1 \bar{b}_1 + a_0 \bar{b}_1 \bar{b}_0 + a_1 a_0 \bar{b}_0$$

a ₁ a ₀ \ b ₁ b ₀		b ₁ b ₀			
		00	01	11	10
00					
01		1			
11		1	1		1
10		1	1		

$$f_{a<b} = \sum m(1, 2, 3, 6, 7, 11)$$

$$= \bar{a}_1 b_1 + \bar{a}_1 \bar{a}_0 b_0 + \bar{a}_0 b_1 b_0$$

a ₁ a ₀ \ b ₁ b ₀		b ₁ b ₀			
		00	01	11	10
00			1	1	1
01				1	1
11					
10				1	